E-commerce Sales Analysis

Portfolio Project Report | Excel • SQL • Power BI • Python

1. Project Overview

This project demonstrates an end-to-end business analytics workflow using order-level e-commerce data. The analysis focuses on revenue, profit, monthly trends, product categories, regions, sales channels, and order status.

Data transparency: The dataset contains 500 synthetic sample records created for demonstration. It is not real company/customer data.

2. Business Questions

• What are revenue, profit, units sold, and profit margin for delivered orders?

• How do revenue and profit vary month by month?

• Which categories and products contribute the most revenue?

• Which regions and sales channels perform best?

• What proportion of records are delivered, returned, or cancelled?

• How should discounts be evaluated against profit?

3. Executive KPIs (Delivered Orders)

Metric	Result
Delivered orders	438
Units sold	1,107
Revenue	INR 1,645,899.55
Profit	INR 494,586.00
Profit margin	30.05%

4. Tools and Workflow

Excel: Executive KPI summary and monthly, category, regional, channel, top-product, and raw-data sheets.
SQL: Queries for KPI summaries, monthly trends, category performance, top products, regional/channel comparisons, order-status distribution, and discount analysis.
Power BI: Dashboard guide with KPI cards, trend chart, category/region/channel comparisons, slicers, and DAX measures.
Python: Pandas aggregations, summary CSV exports, and monthly revenue chart generation.

5. Business Recommendations

• Prioritize marketing and inventory planning for high-performing categories after comparing their profit margins.

• Compare regional results and test targeted campaigns in lower-performing regions.

• Evaluate discounts using profit as well as revenue to avoid unprofitable growth.

• Track returned and cancelled orders separately and investigate patterns by category.

• Refresh the dashboard monthly and validate data quality before making decisions.

6. Limitations

This is a demonstration project built on synthetic data. Results are illustrative and should not be treated as actual market evidence. A Power BI Desktop report (.pbix) must be created by importing the supplied CSV and following the included Power BI dashboard guide.

7. Suggested Interview Explanation

"I created an E-commerce Sales Analysis project to demonstrate how raw order data can be turned into business insights. I used Excel for summary analysis, wrote SQL queries for key business questions, prepared a Power BI dashboard plan with KPI cards and filters, and added Python code to reproduce the main summaries. The dataset is synthetic and is used only to demonstrate my workflow."